DEEPIKA D

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Professional Summary

Aspiring VLSI and Embedded IoT Engineer with a strong foundation in Digital Electronics, Analog Electronics, CMOS Technology, MOSFET Fundamentals, Verilog HDL, and VHDL. Hands-on experience in RTL Design, Functional Verification, FPGA-based projects, and IoT application development. Familiar with VLSI design flow, simulation, synthesis, and embedded system development. Seeking opportunities in Semiconductor Design, Verification, and Embedded Engineering.

Education

B.E. Electronics and Communication Engineering — **2023 – 2027**
KSR Institute for Engineering and Technology — **CGPA: 8.95**

Higher Secondary School (HSS) — **2022 – 2023**
Hillfort Matric and Higher Secondary School — **Percentage: 79%**

Technical Skills

HDL	Verilog HDL, VHDL
VLSI Concepts	CMOS Technology, MOSFET Fundamentals, RTL Design, FSM Design, Combinational Logic, Sequential Logic, Timing Analysis
Embedded & IoT	Arduino, Embedded C, Sensors, IoT System Design
Programming	C, Python
Tools	FPGA Tools, Simulation Tools

Internships

VLSI Intern — **Caliber Embedded Technologies, Coimbatore**

- Developed RTL modules using Verilog HDL including adders, counters, and multiplexers.
- Performed functional simulation and verification of digital circuits.
- Gained exposure to synthesis flow and timing analysis.

IoT Intern — **Samecore Solutions, Trichy**

- Worked on IoT-based automation and monitoring applications.
- Integrated sensors and microcontrollers for real-time data acquisition.
- Developed embedded solutions using Arduino and IoT communication modules.

Projects

Traffic Light Controller using VHDL

- Designed and implemented an FSM-based traffic light controller using VHDL.
- Performed simulation and functional verification using FPGA tools.
- Ensured correct state transitions and timing control.

Smart Fire Detection System (Embedded IoT)

- Developed an IoT-based fire detection system using flame and gas sensors.
- Implemented real-time monitoring and emergency alert mechanisms.
- Enhanced safety through instant hazard detection and notification.

Smart Lock System (Embedded IoT)

- Designed an IoT-enabled smart lock system for secure access control.
- Integrated authentication and remote monitoring features.
- Improved security through automated access management.

Certifications

- Functional Verification using SystemVerilog and UVM
- RTL Design using Verilog HDL
- NPTEL Certification – Internet of Things (IoT)
- VLSI Internship Certificate – Caliber Embedded Technologies
- IoT Internship Certificate – Samecore Solutions, Trichy

Soft Skills

• Time management • Adaptability • Team Collaboration